Case Study: Analyzing Customer Churn in Power BI

Project Scope:

- Analyze customer churn data in Power BI to identify key patterns and factors driving customer attrition through exploratory analysis and measure creation.

- Perform data modeling, create calculated measures, and design interactive dashboards to uncover actionable insights and trends in customer behavior.

- Develop clear, story-driven dashboard visuals to communicate churn insights and support data-driven decision-making.

Senior	Under 30	Churn Rate
No	No	24.54 %
No	Yes	23.00 %
Yes	No	38.46 %
Total		**26.86 %**

6687

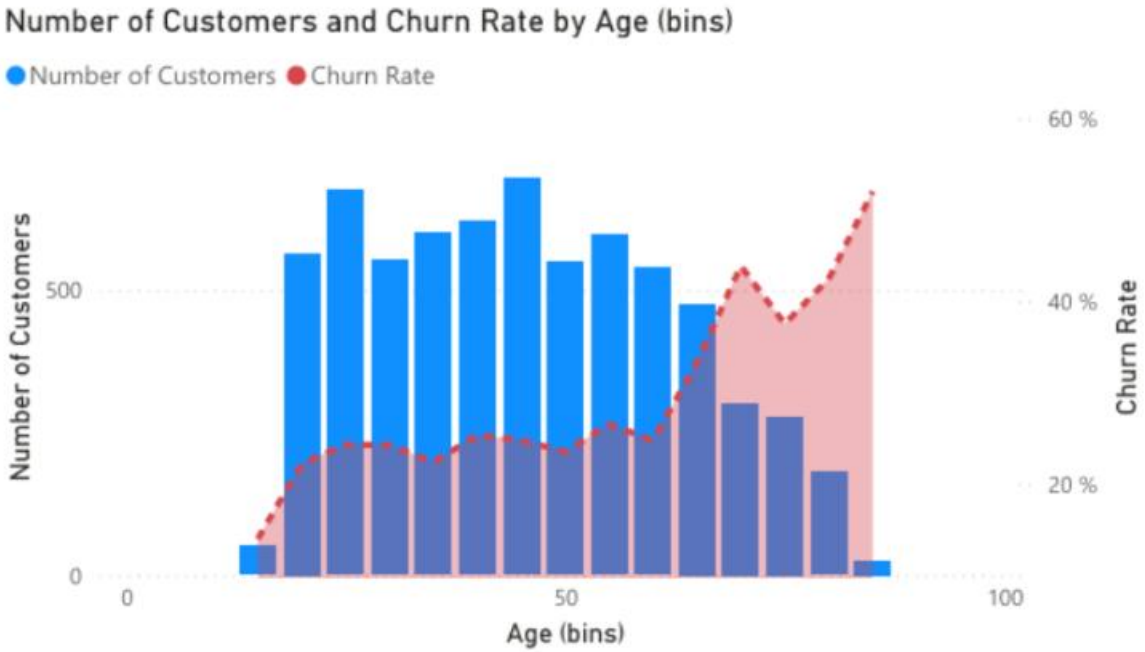
Count of Customer ID

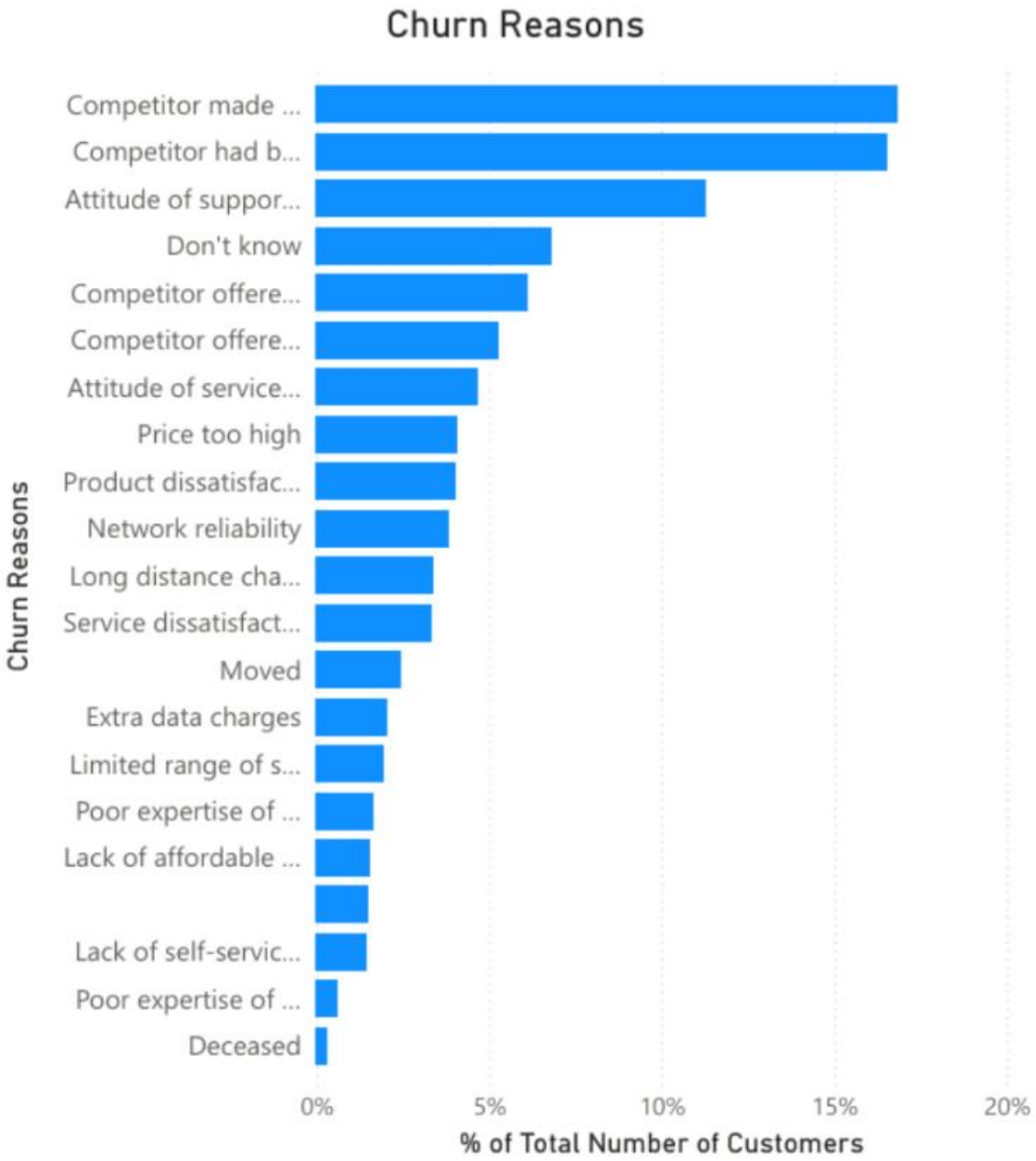
26.86 %

Churn Rate

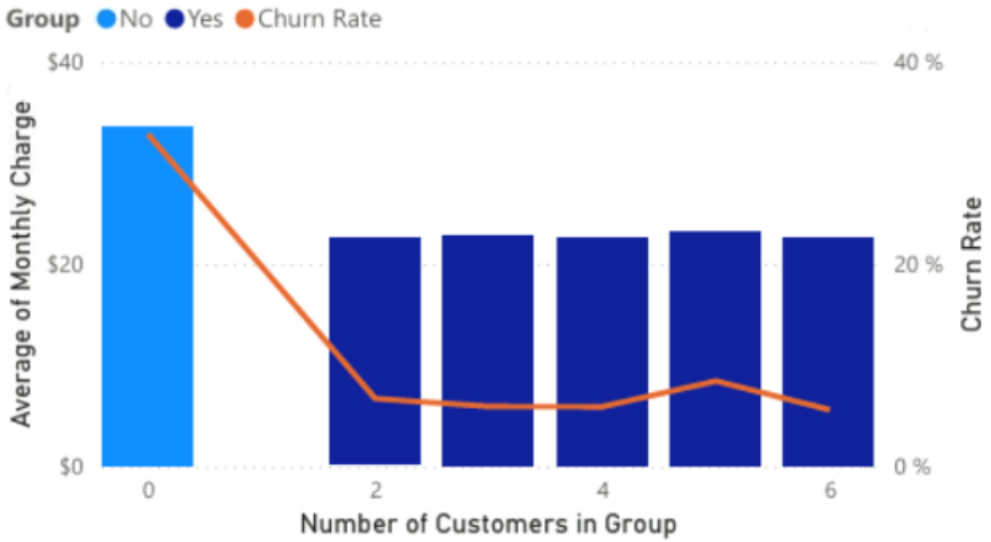
1796

Churned

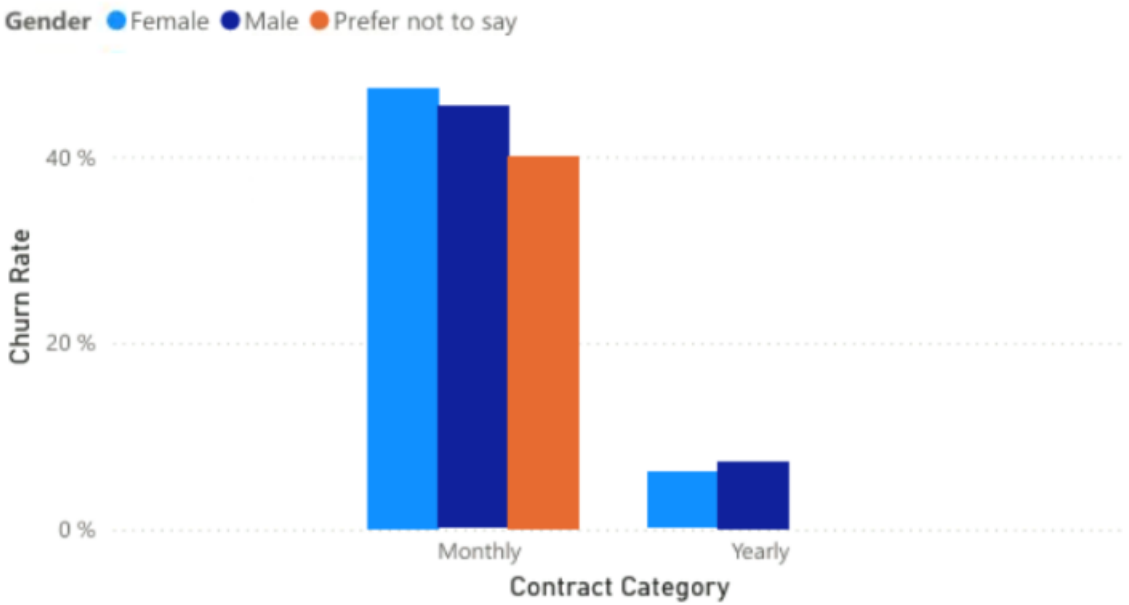




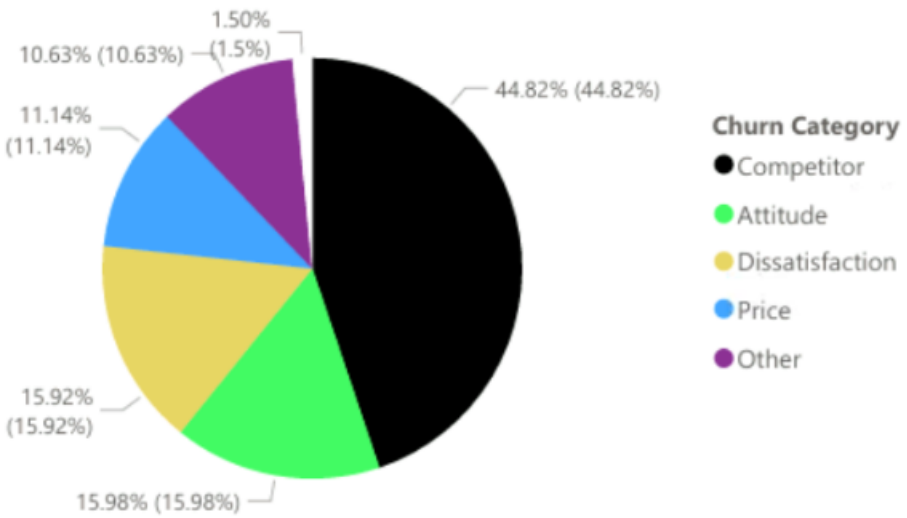
Average of Monthly Charge and Churn Rate by Number of Customers in Group and Group



Churn Rate by Contract Category and Gender



Churn by category



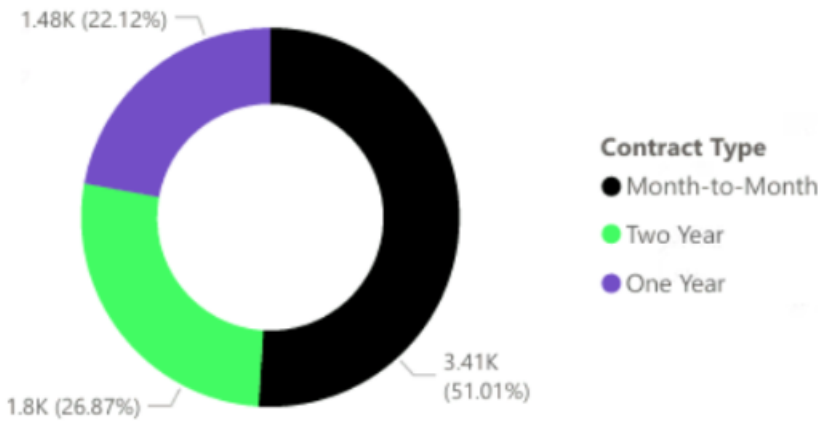
Monthly

46.29 %
Churn Rate

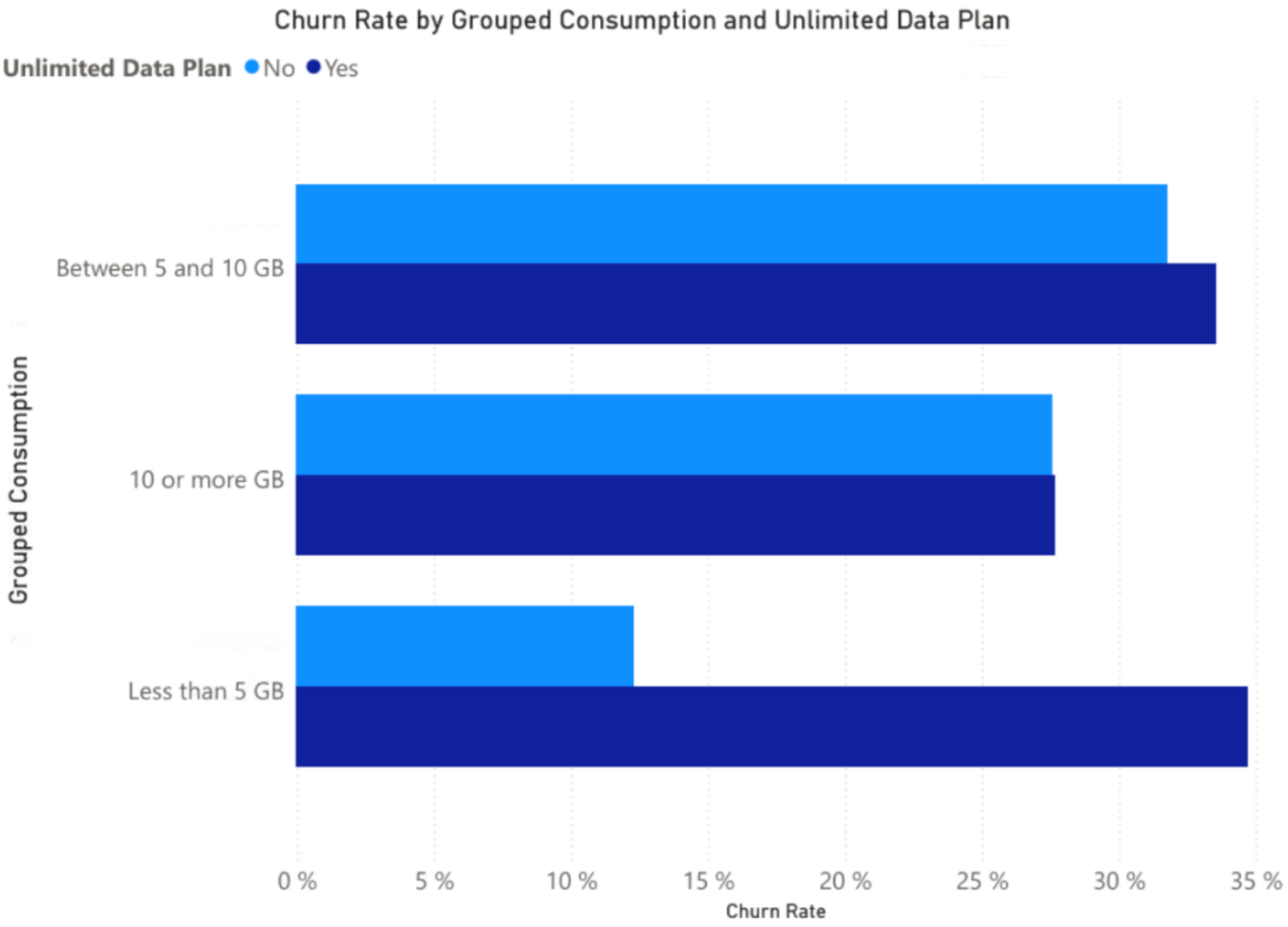
Yearly

6.62 %
Churn Rate

Customers by Contract type

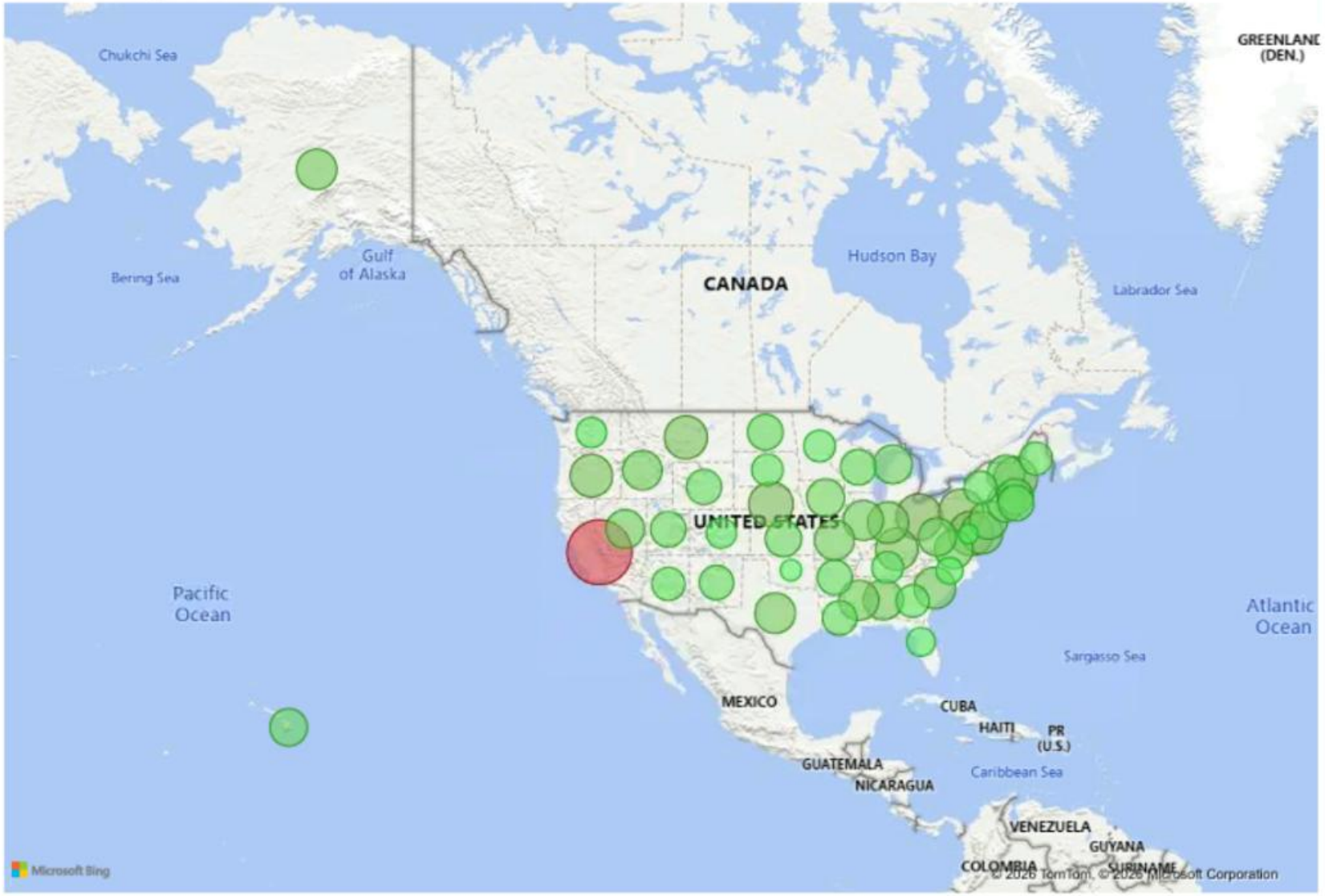


Churn Rate	Number of Customers	Unlimited Data Plan
16.10 %	2193	No
32.11 %	4494	Yes
26.86 %	**6687**	



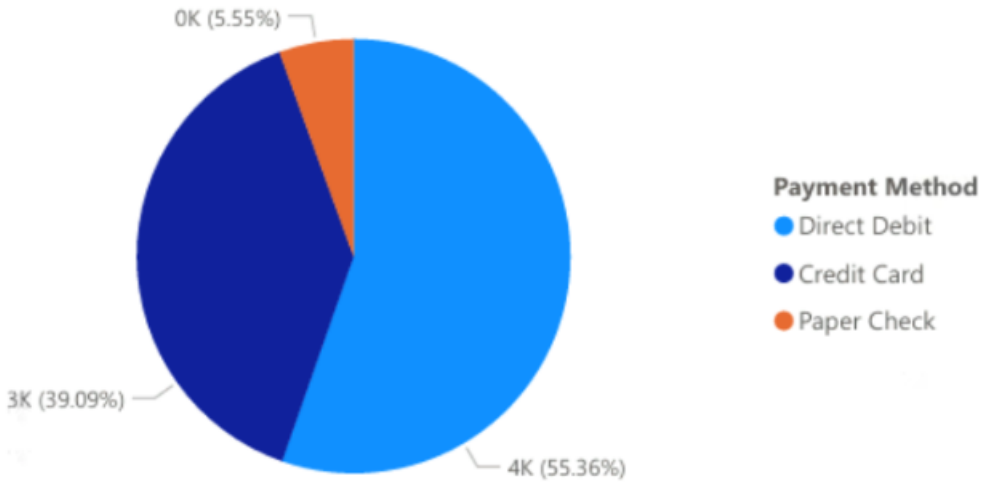
Intl Active	no	yes	Total
No	20.01 %	71.19 %	22.21 %
Yes	40.34 %	7.59 %	34.31 %
Total	**27.07 %**	**24.88 %**	**26.86 %**

Churn Rate by State

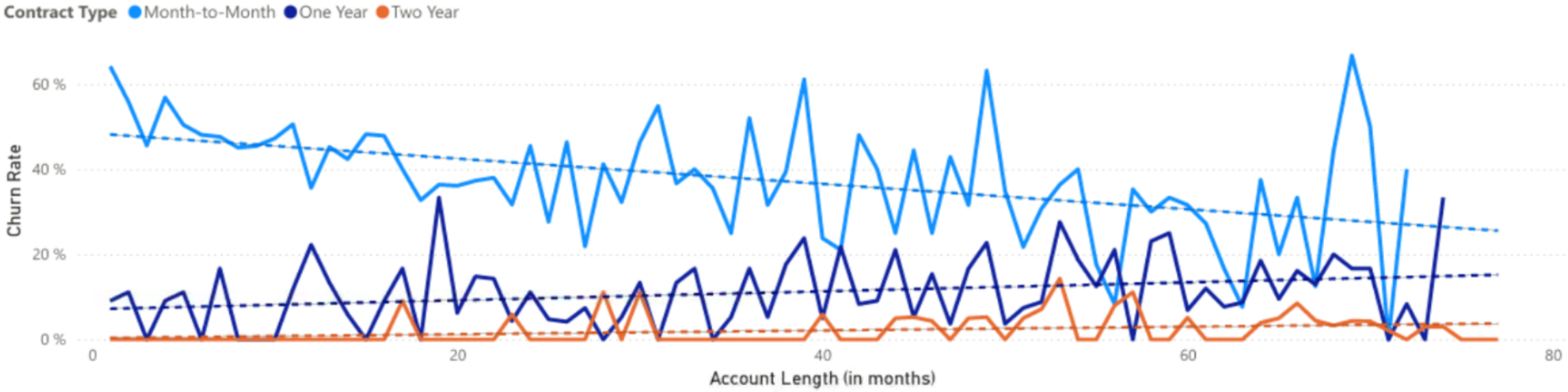


Churn Rate by Account Length (in months)

Number of Customers and Churn Rate by Payment Method



Churn Rate by Account Length (in months) and Contract Type



26.86 %
Churn Rate

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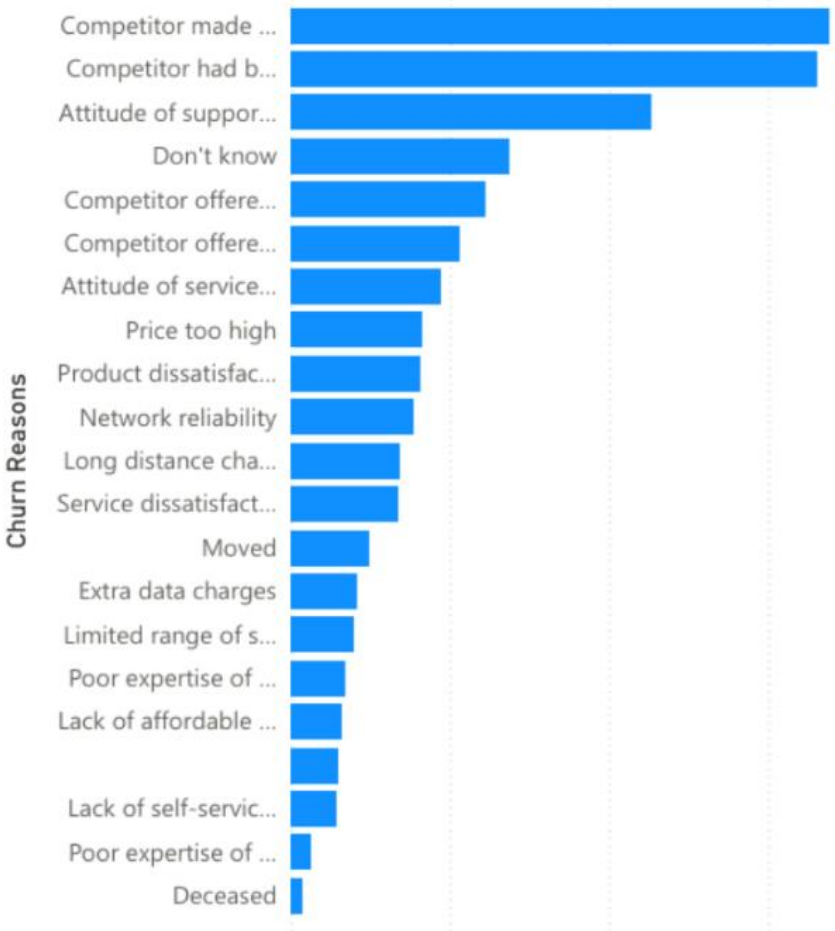
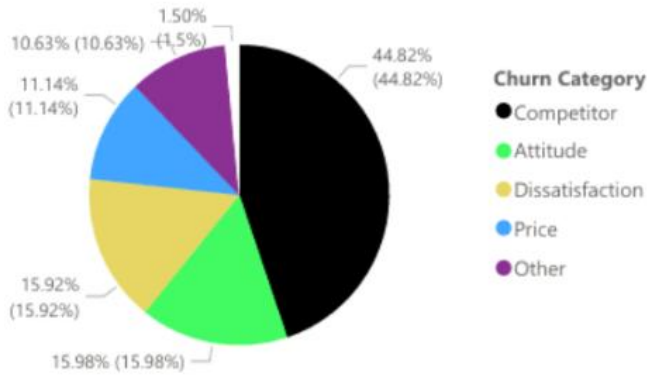
Churned

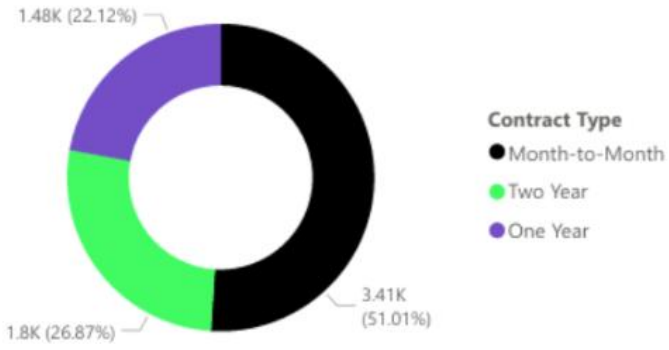
Churn Reasons



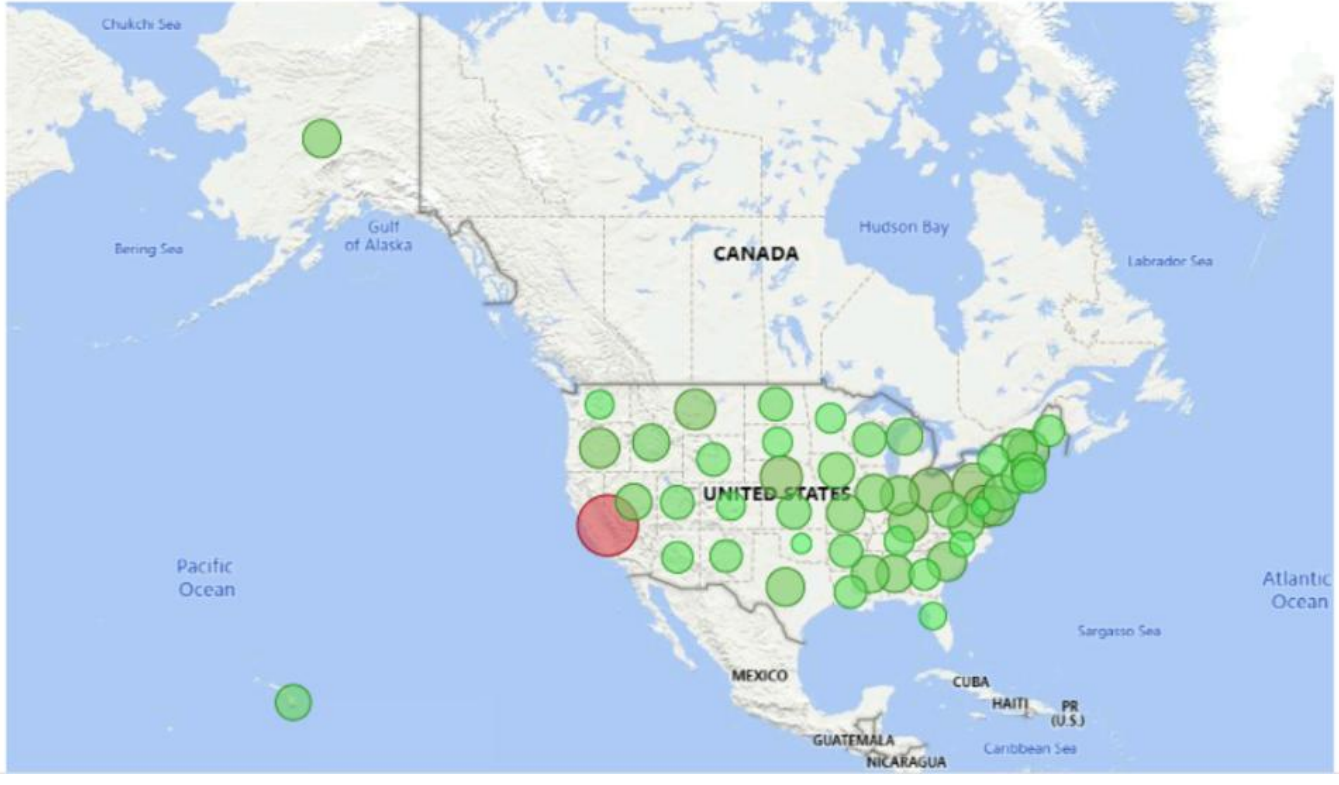
%GT Count of Churn Label by Churn Category

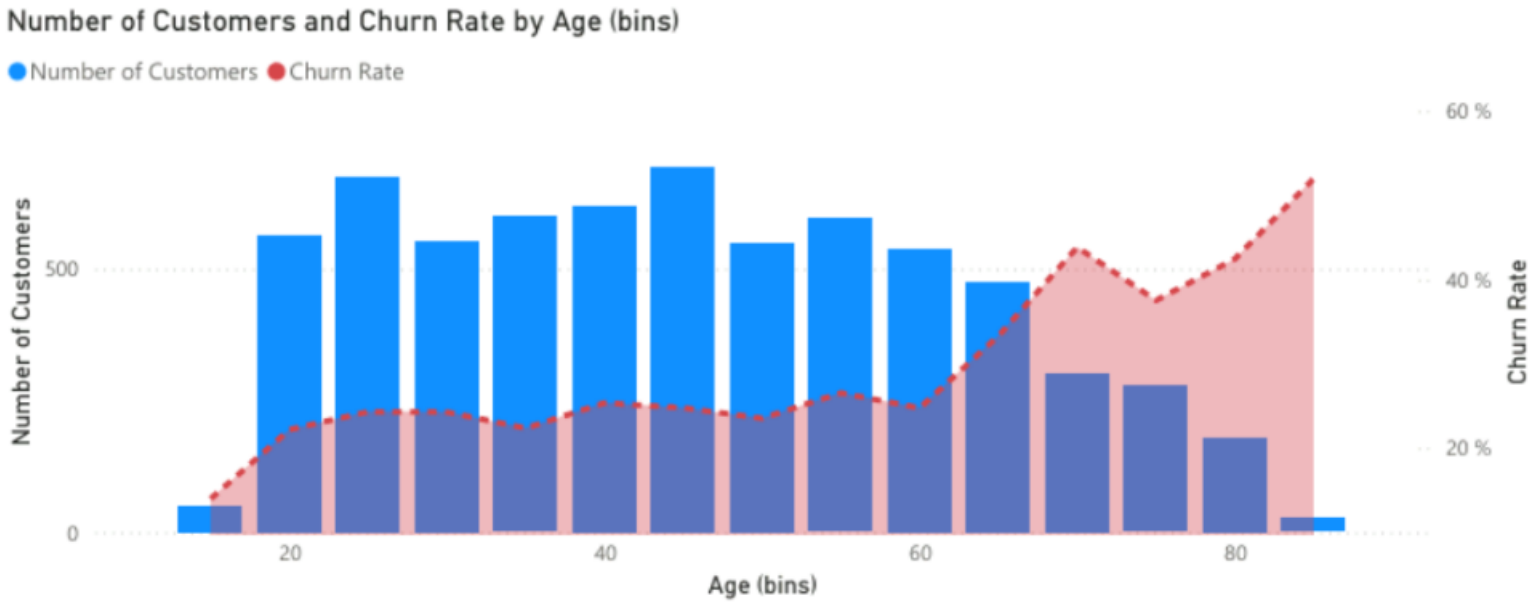


Count of Customer ID by Contract Type



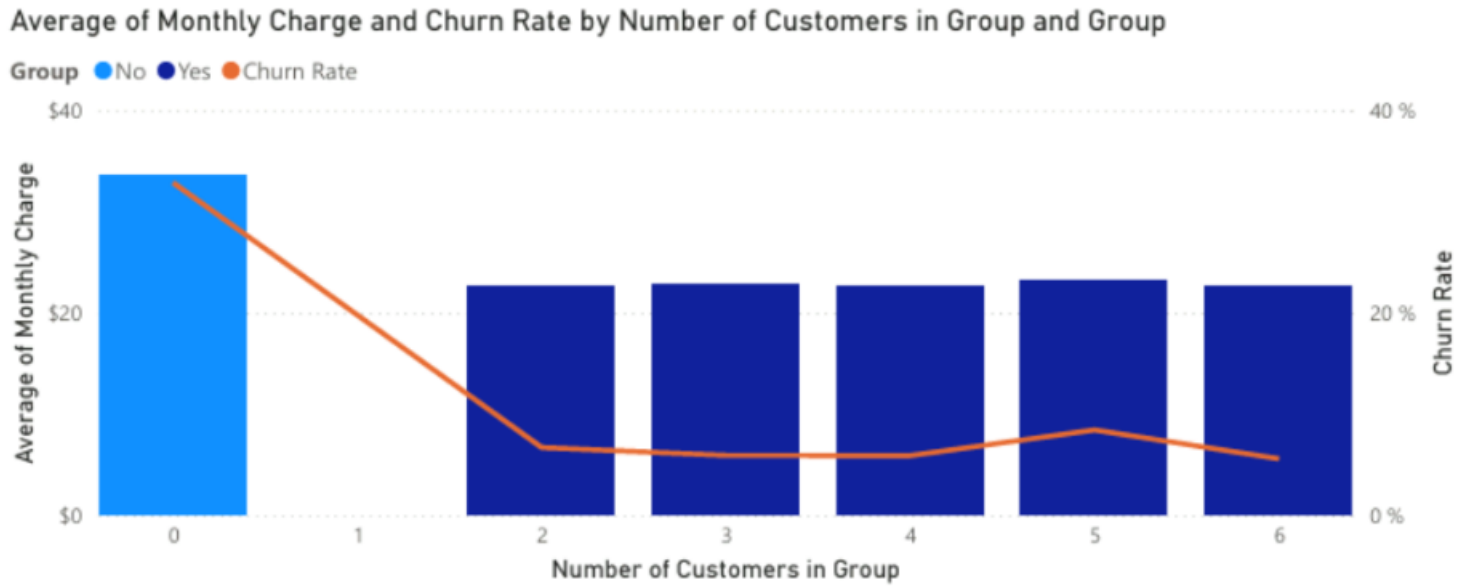
Churn Rate by State





Account Length (in months)

1 77

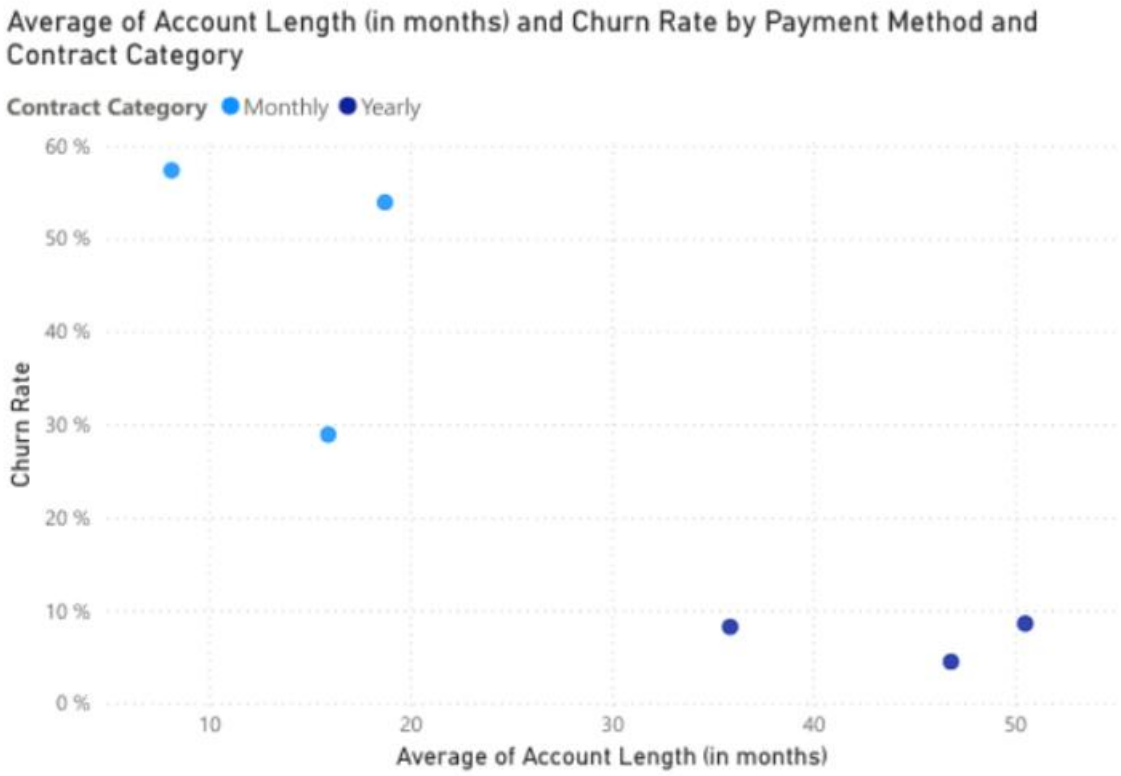


6123

Customer Service Calls

0.92

Avg Customer Service Calls



Contract Category, Payment Method

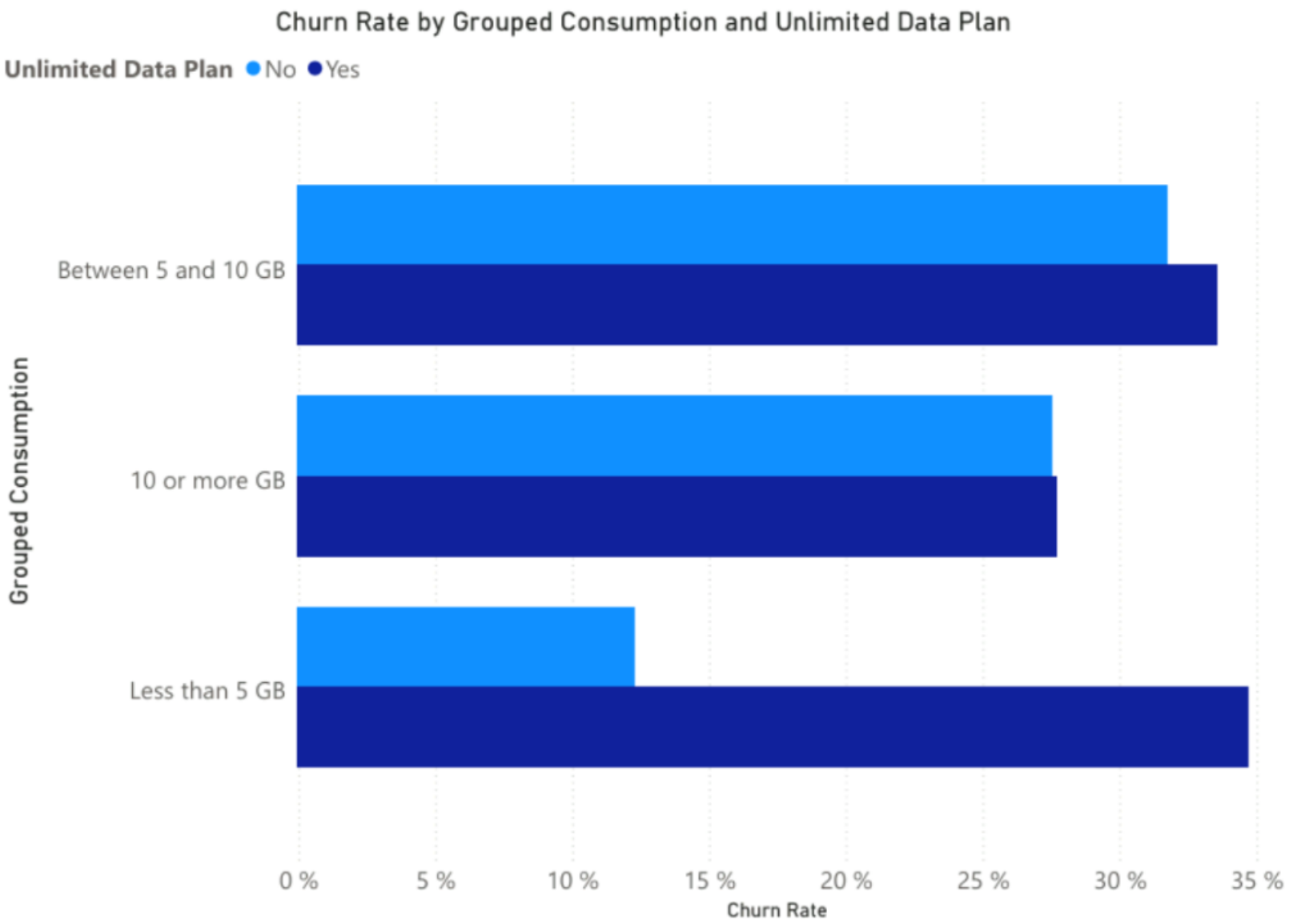
- Monthly
  - Credit Card
  - Direct Debit
  - Paper Check
- Yearly
  - Credit Card
  - Direct Debit
  - Paper Check

33.64

Avg Extra International Charges

3.37

Avg Extra Data Charges



6123

Customer Service Calls

33.64

Avg Extra International Charges

0.92

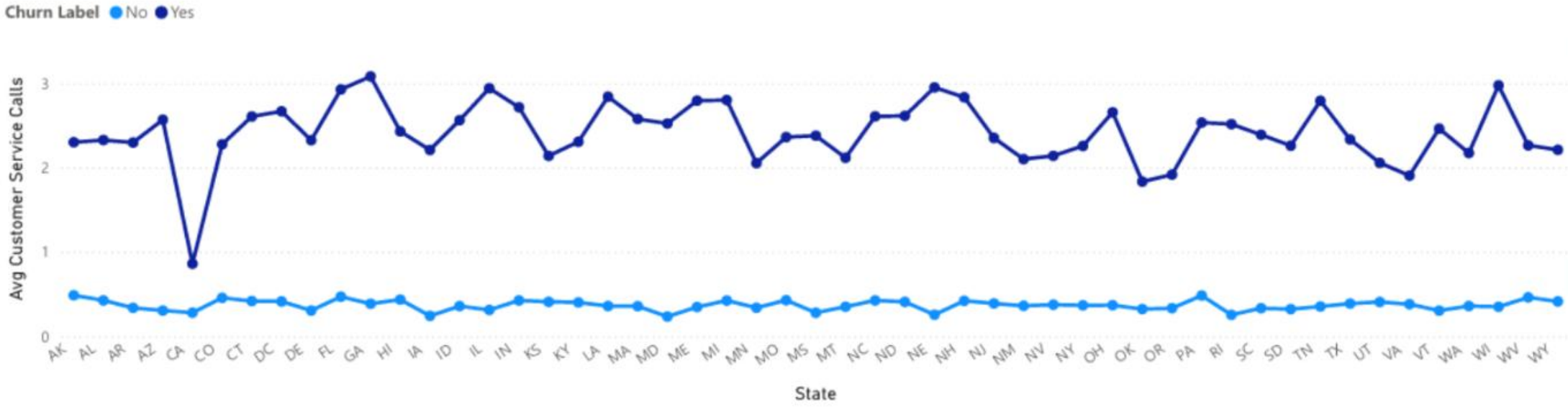
Avg Customer Service C...

3.37

Avg Extra Data Charges

Churn Rate by State

Avg Customer Service Calls by State and Churn Label



Key Insights

Churn Overview

- 26.86% overall churn rate — 1,796 of 6,687 customers lost
- 44.8% of churners left because of a competitor — the #1 reason by far
- 38.5% churn rate for senior customers — nearly 2x the average
- Attitude/support issues drive 15.98% of churn — a fixable service quality gap

Contract & Loyalty

- Monthly contract churn is 46.3% — 7x higher than yearly contracts at just 6.6%
- 51% of all customers are on Month-to-Month — the single highest-risk segment
- Group plan churn drops sharply as group size grows — a strong loyalty protection effect
- Churn declines from ~60% at month 1 down to near 0% by month 72 — tenure = retention

Service & Geography

- 71.2% churn for international plan users who are not actively making calls — a pricing mismatch
- 32.1% churn for unlimited data plan holders — plan value is unclear to customers who use <5GB
- Churners average 2-3x more customer service calls — a leading warning signal
- $33.64 avg extra international charges — a key friction point pushing customers out
- One state is flagged as a high-risk outlier on the map — a geographic priority to investigate (There's one more, orange in color, worth noticing)

Recommendation:

- Convert monthly users to annual plans · Win back senior customers · Fix international plan pricing · Improve support quality to reduce attitude-driven churn